

AMANDA L. SKARLUPKA, PhD, MS

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PROFESSIONAL SUMMARY

Senior program manager and PhD data scientist who builds the operating backbone for complex, multi-stakeholder research and data programs. Ten years across federal government (NIH/NCI), nonprofit research infrastructure, and academic research, covering portfolio and grants management, AI/ML governance, data sharing policy, and strategic planning. Turns ambiguous technical programs into documented, metrics-driven operations, and translates fluently between scientists, engineers, policy staff, and executive leadership.

CORE COMPETENCIES

Program & Portfolio Management (42 concurrent projects; Jira) · Federal Grants, Contracts & Interagency Agreements (FAC-COR Level 1) · Data Governance & Sharing Policy (FAIR/CARE/TRUST, EHDS) · AI/ML Governance & Evaluation · Strategic Planning & Impact Assessment · Stakeholder & Partner Engagement (WHO, NIH, industry) · Technical Writing & Executive Briefing · R, Python, reproducible analysis

PROFESSIONAL EXPERIENCE

Senior Program Manager | *Vivli, Inc.*

April 2025 – Present

- Lead program management infrastructure for 42 concurrent projects; established documentation standards, Jira-based workflow, and metrics-driven reporting to support data-informed decision-making.
- Serve as subject matter expert in data governance and international data sharing policy, advising on European Health Data Space (EHDS) regulations, U.S. DOJ rulings on countries of concern, and federal agency expectations.
- Oversee the AMR Data Challenge and AMR Register operations including member recruitment, stakeholder engagement, and strategic partnerships with WHO, The Global Health Network, and AMR Insights.
- Contribute to NIH's Generalist Repository Ecosystem Initiative (GREI) on AI metadata curation, data standards, and metrics development; lead development of impact assessment framework to prioritize organizational initiatives.
- Author scientific manuscripts and white papers to advance FAIR data sharing and Vivli's strategic mission; presented at ESCMID Global 2026 and NCI Office of Data Sharing Symposium 2025.

Health Scientist Administrator – Program Officer | *National Cancer Institute*

June 2024 – April 2025

Division of Cancer Prevention, Office of the Director

- Managed a grants/contracts portfolio in data science, AI/ML governance, precision medicine, and ethics; applied knowledge of NIH/NCI strategic directions to evaluate portfolio gaps and recommend programmatic changes.
- Provided regulatory and technical expertise supporting AI-enabled device approvals for clinical utility trials; synthesized advances across AI, imaging, and oncology into policy recommendations for agency leadership.
- Served as Executive Secretary, NIH GREI Council of Councils Working Group; represented DCP on NCI-wide AI Working Group, NIH Indigenous Data Sovereignty Working Group, NCI-DOE Concept Tiger Team, and PRIME-AI Common Fund Working Group.
- Managed federal contracts (Lung Cancer Screening Image Library), clinical trial agreements (Cancer Screening Research Network), and interagency agreements (NASA/JPL); maintained FAC-COR Level 1 certification.
- Conceptualized and published Notices of Funding Opportunity (NOFOs/NOSIs); reviewed proposals for NCATS, ODSS, and NCI programs involving AI, federated learning, digital twins, and data reuse.
- Prepared substantive briefings, reports, and findings for NCI/DCP leadership; represented DCP perspectives at NIH-wide and inter-governmental committees and working groups.

Presidential Management Fellow – Health Specialist | *National Institutes of Health*

June 2022 – June 2024

NCI Division of Cancer Prevention; Office of Data Science Strategy (detail)

- Designed data collection and management frameworks for the Multi-Cancer Detection (MCD) clinical network to optimize data sharing and reuse; advised on AI/ML best practices for public health research.
- Drafted a Notice of Special Interest (NOSI) on NIH data reuse; conducted reproducible portfolio analysis in R evaluating institutional research investments in secondary data analysis (ODSS rotation).
- Coordinated the NIH Strategic Plan mid-course review data call across ODSS; authored the AI Bold Prediction incorporated into the NIH-wide Strategic Plan ("AI will reveal molecular signatures associated with the return to health after an acute illness").
- Organized the Data Repository and Knowledgebase PI Meeting and contributed to DS3C and NIH Interagency Modeling and Analysis Working Groups. Co-led ATC DEIA-C Working Group; chaired Education Subcommittee establishing a monthly IC-wide seminar series.

Doctoral Graduate Researcher | *University of Georgia* | *NIH CIVR-HRP*

January 2018 – May 2022

- Led 4 concurrent funded research projects on broadly-protective influenza vaccine development; managed 4 collaborating research groups; mentored 3 undergraduate researchers; secured \$21,000 in dissertation funding.
- Applied COBRA computational vaccine design, multidimensional statistical analyses, and cross-disciplinary methods (virology, immunology, biostatistics); authored 15+ peer-reviewed publications and contributed to 2 NIH grant submissions. Holds PCT patent (PCT/US2021/012695).

ADDITIONAL EXPERIENCE

Master's Graduate Researcher | University of Georgia, College of Public Health | 2019 – 2022 — Quantified influenza vaccine antibody breadth using multidimensional scaling and PCA in R; harmonized public and unpublished datasets; earned the M.S. concurrent with the Ph.D.

Associate Research Specialist | Food Research Institute, UW–Madison | 2014 – 2016 — Designed and ran pathogen growth and thermal inactivation studies for industry clients; authored protocols, analyzed results, briefed clients, supervised students; Tier 1 Federal Select Agent Program approved.

Undergraduate Researcher | Department of Bacteriology, UW–Madison | 2013 – 2014 — Molecular genetics of *Aspergillus* spp.: gene-deletion strain construction and transcriptional expression analysis.

EDUCATION

Ph.D., Infectious Disease | University of Georgia | May 2022

M.S., Epidemiology & Biostatistics | University of Georgia | May 2022

B.S., Microbiology & Botany | University of Wisconsin–Madison | May 2014

AWARDS & RECOGNITION

- NIH Office of the Director Honor Award – Emerging Leader (2024)
- NCI Director's Award: NCI Artificial Intelligence Working Group (2024)
- NCI Director's Award: Vanguard Study Assay Verification & Selection Working Group (2023 & 2024)
- Presidential Management Fellow, NIH Class of 2024 | Dissertation Completion Award (\$21,000), UGA

SELECT COMMITTEES & WORKING GROUPS (WG)

NCI-Wide AI WG · PRIME-AI Common Fund WG · GREI Council of Councils WG (Exec. Secretary) · MCD Clinical Research Network Tiger Team · NCI-DOE Concept Tiger Team · NIH-Wide DS3C WG · NIH Interagency Modeling & Analysis WG

SELECT PUBLICATIONS (21 peer-reviewed; PCT patent holder; full list available)

LeeVan* & Skarlupka* et al. (2025). Framework to Select Multi-Cancer Detection Assays in NCI's Vanguard Study. *Cancer Epidemiol Biomarkers Prev.* DOI: 10.1158/1055-9965.EPI-25-0903

Wood, Pinsky, ..., Skarlupka et al. (2026). Performance of multiple multi-cancer detection tests using a large independent reference set. *J Natl Cancer Inst.* DOI: 10.1093/jnci/djag001

Billings, Ge, Skarlupka et al. (2025). Different antigenic distance metrics generate similar predictions of influenza vaccine response breadth. *PLoS Comput Biol.* DOI: 10.1371/journal.pcbi.1013720

Rubinstein* & Patriotis* ... Skarlupka et al. (2024). Cancer Screening with Multi-Cancer Detection Tests: A Translational Science Review. *CA Cancer J Clin.* DOI: 10.3322/caac.21833

PROFESSIONAL AFFILIATIONS & SERVICE

ESCMID · American Society for Microbiology · AIM-AHEAD Consortium | Peer reviewer: PLOS Computational Biology, *Frontiers in Immunology*, *BMC Microbiology* | Judge: NIH/FASEB HeroX DataWorks! Challenge; Georgia Junior Science & Humanities Symposium